AHMAD ARI ALDINO

LEARNING ANALYTICS | AI IN EDUCATION | GENERATIVE AI & LLM EVALUATION

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Professional Profile

PhD candidate in Information Technology (Learning Analytics) at Monash University, researching how AI can support learner-centred feedback and student learning in higher education. My work combines generative AI and LLM evaluation, educational NLP, machine learning, authentic assessment data, and quantitative and text-analytic methods. I have first-author publications in leading educational research journals, including *Computers & Education*, *Computers and Education: Artificial Intelligence*, and *Assessment & Evaluation in Higher Education*, alongside university teaching experience in Australia and Indonesia.

Research Expertise

- **Learner-centred feedback and learning sciences:** Turned an established feedback framework into measurable categories: does feedback guide students' next tasks (future impact), help them understand their current work (sensemaking), and encourage them to take charge of their learning (agency)?
- **Generative AI and LLM evaluation:** Designed and evaluated LLM-based feedback workflows across prompting conditions; compared AI-generated and teacher feedback on usefulness for learning and writing quality.
- **Educational NLP and dataset curation:** Curated and annotated authentic higher-education feedback and student-work datasets; applied supervised and multilabel classification, BERT-based modelling, and validation procedures.
- **Learning analytics and student outcomes:** Applied statistical and text-analytic methods to feedback, educator–AI interaction, student revision actions, and subsequent academic performance.

Doctoral Research

Five connected studies on learner-centred feedback, AI-supported analysis, educator–AI interaction, and outcomes.

- **Study 1 | Mapping learner-centred feedback.** Analysed authentic higher-education feedback, its alignment with learner-centred components, and patterns across student groups. **Published in *Computers & Education* (2025).**
- **Study 2 | Automating feedback analysis.** Developed and evaluated machine learning and NLP approaches to classify learner-centred feedback; BERT-based classifiers achieved Cohen's κ up to 0.956 and F1 up to 0.998. **Published in *Computers and Education: Artificial Intelligence* (2024);** implemented in PolyFeed.
- **Study 3 | Human and LLM-generated feedback.** Analysed 5,156 human- and LLM-generated feedback texts on 1,289 student proposals, comparing how useful the feedback is for learning and how it is written, across prompting strategies. **Under review.**
- **Study 4 | Teacher–AI collaboration.** Investigated educators' interaction with and perceptions of AI-supported feedback in PolyFeed, focusing on use, revision, trust, ethics, and usability. **Published in *Assessment & Evaluation in Higher Education* (2026).**
- **Study 5 | Feedback and student outcomes.** Analysed assessment and feedback data from 1,929 students across three teaching semesters to examine associations between learner-centred feedback components, student revision actions, and subsequent academic performance. Manuscript in preparation.

Research Output

- **PolyFeed | AI-supported feedback analytics system.** Includes the learner-centred feedback classifier I built in Study 2, used in my research on automated feedback analysis and educators' use of AI-supported feedback. PolyFeed team

Academic Appointments

- **Teaching Associate** | Monash University, Faculty of Information Technology — Feb 2023 – Present
  Teach postgraduate data science, including FIT5145 since 2023. Facilitate tutorials and consultations, assess assignments and examinations, and give constructive feedback to support learning.
- **Lecturer (Assistant Professor)** | Universitas Teknokrat Indonesia — Sep 2019 – Present
  Teach undergraduate data mining, statistics, data visualisation, data warehousing, and data science. Design syllabi, learning objectives, and assessments; supervise student research and theses; contribute to curriculum development.
- **Assistant Lecturer** | Universitas Lampung — Sep 2013 – Sep 2015

Education

- **PhD, Information Technology (Learning Analytics)** | Monash University, Australia — Jan 2023 – Present
  Centre for Learning Analytics · Beasiswa Pendidikan Indonesia (BPI) scholarship, 2022 · Thesis submission: Dec 2026

- **Master of Mathematics** | Universitas Lampung, Indonesia — Sep 2017 – Apr 2019
  Thesis: Study on Metric Dimensions on a Graph and Its Applications.
- **Bachelor of Mathematics** | Universitas Lampung, Indonesia — Sep 2012 – Dec 2015

Technical Skills and Research Methods

- **Programming and analytics:** Python, R, SQL, SPSS, NVivo, Tableau, Excel, MATLAB, Unix/Bash, LaTeX.
- **AI and NLP:** BERT, XGBoost, random forest, supervised and multilabel classification, LLM API workflows, prompt design, text analytics.
- **Research methods:** Mixed-effects and count models, equivalence testing, randomised and quasi-experimental designs, qualitative thematic analysis, learning analytics, dataset annotation, model validation.

Academic Service

Member, Society for Learning Analytics Research (SoLAR), 2024–present. **Peer reviewer (2024–2026):** *Computers & Education: Artificial Intelligence*; *Teaching and Teacher Education*; *Education and Information Technologies*; *Assessment & Evaluation in Higher Education*; *Heliyon*; *Cogent Business & Management*.

Publications

Google Scholar (November 2026): 1,016 citations · h-index 13 · i10-index 15

Selected journal articles: learning analytics and AI in education

- **Aldino, A. A.**, Maheshi, B., Li, Y., Zhou, Y., Tsai, Y.-S., Gašević, D., & Chen, G. (2026). Enhancing learner-centered feedback with AI: Teachers' practices and perceptions. *Assessment & Evaluation in Higher Education, 51*(5), 892–910. doi.org/10.1080/02602938.2026.2638920
- **Aldino, A. A.**, Tsai, Y.-S., Gupte, S., Henderson, M., Nath, D., Gašević, D., & Chen, G. (2025). Analytics of learner-centered feedback: A large-scale case study in higher education. *Computers & Education, 237*, 105360. doi.org/10.1016/j.compedu.2025.105360
- Dai, W., Cheng, Y., **Aldino, A. A.**, Tsai, Y.-S., Gašević, D., & Chen, G. (2025). Evaluating the capability of large language models in characterising relational feedback: A comparative analysis of prompting strategies. *Computers and Education: Artificial Intelligence, 8*, 100427. doi.org/10.1016/j.caeai.2025.100427
- **Aldino, A. A.**, Tsai, Y.-S., Mello, R. F., Gašević, D., & Chen, G. (2024). Enhancing feedback quality at scale: Leveraging machine learning for learner-centered feedback. *Computers and Education: Artificial Intelligence, 7*, 100332. doi.org/10.1016/j.caeai.2024.100332
- Dai, W., Tsai, Y.-S., Lin, J., **Aldino, A. A.**, Jin, H., Li, T., Gašević, D., & Chen, G. (2024). Assessing the proficiency of large language models in automatic feedback generation: An evaluation study. *Computers and Education: Artificial Intelligence, 7*, 100299. doi.org/10.1016/j.caeai.2024.100299

Manuscripts under review and in preparation

- **Aldino, A. A.**, Tsai, Y.-S., Gašević, D., & Chen, G. Can AI be learner-centred? Comparing human and generative AI feedback in higher education. *Under review.* (Doctoral Study 3)
- Rataphaibul, P.*, **Aldino, A. A.***, Nawaz, S., Maheshi, B., Jin, F., Chen, G., & Tsai, Y.-S. Exploring students' feedback literacy: A study of differences in feedback interactions based on confidence and performance expectation alignment. *Under review, Journal of Learning Analytics.* (*Joint first authors)
- Maheshi, B., **Aldino, A.**, Swiecki, Z., Lim, E., Nikfalazar, S., Martinez-Maldonado, R., & Tsai, Y.-S. Making feedback traceable: Examining how students act on feedback. *Under review.*
- Maheshi, B., Palihena, H., **Aldino, A.**, Jin, F., Wang, Z., Ng, K. H., Gašević, D., Chen, G., Charwat, N., Chan, P. W. K., Nawaz, S., Martinez-Maldonado, R., & Tsai, Y.-S. How do students interact with my feedback? A feedback analytics tool to facilitate dialogic feedback. *Under review, Journal of Computer Assisted Learning.*
- Maheshi, B., Wang, Z., **Aldino, A.**, Palihena, H., Ng, K. H., Martinez-Maldonado, R., & Tsai, Y.-S. Facilitating dialogic feedback using student-facing and educator-facing feedback analytics dashboards. *Under review, Journal of Learning Analytics.*
- Liang, Z., Li, Y., **Aldino, A.**, Sha, L., Gašević, D., & Chen, G. GenAI feedback for supporting students' goal setting and enactment: A mixed-methods study. *Under review, International Journal of Human–Computer Interaction.*
- **Aldino, A. A.**, Tsai, Y.-S., Gašević, D., & Chen, G. Does learner-centred feedback pay off? Evidence from student uptake and performance change in higher education. *In preparation.* (Doctoral Study 5)

Other education-related publications

- Fang, Z., Han, J. Y., **Aldino, A. A.**, Feng, Z., & Cai, T. (2024). Decoding the naturally arisen leadership in collaborative learning: A Bayesian simulation and human network study. *Proceedings of the 39th ACM/SIGAPP Symposium on Applied Computing*, 21–28.
- Tusadiyah, H., Anggoro, B. S., **Aldino, A. A.**, & Ambarwati, R. (2024). Puzzle-assisted REACT learning strategy on mathematical literacy and adaptive reasoning ability: A study of Indonesian junior high school students. *Journal of Mathematics and Science Education, 1*(1), 14–24.
- Yeztiani, O. L., Adrian, Q. J., & **Aldino, A. A.** (2022). Application of augmented reality as a learning media of Mollusca group animal recognition and its habitat based on Android. *Jurnal Teknoinfo, 16*(2), 420–426.

19 further publications (2020–2025) in applied data science and decision support; full list on Google Scholar.

Referees

Available on request.